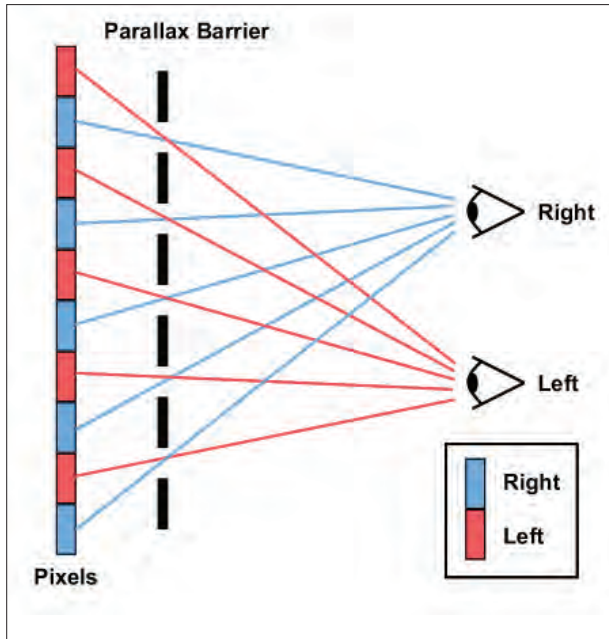


There are two main technologies for auto-stereoscopic displays – parallax barrier and lenticular lens.

Lenticular lens

Lenticular lens displays are very costly, and large in size, so are not suitable for home usage. Lenticules are tiny lenses that are fused on to the rear side of a



special film. This film coating is put on the display's surface. Two sets of the same image are displayed on screen, and the lenses work to direct light to each eye such that each eye only sees one of the images. These lenses are cylindrical in shape, and each lens directs light from alternate pixel segments to the same spot. There are two such spots (obviously!), one for each eye. Your brain interprets this as a single 3D image. One of the disadvantages of such displays is that content needs to be created and optimised for this kind of display. Another problem is that, since light rays and lenses are involved, there is a sweet spot to get the most of the 3D effect. If you move out of this zone, the picture would start to get blurry. The obvious solution would be to include some sort of camera that would track movement and adjust accordingly. One problem would be a scenario where five people are watching one 3D screen where the camera needs to track all of their movements and calculate each sweet

spot – this seems a little elaborate to be of practical use at the moment.

Parallax barrier

The term parallax signifies the apparent difference in position of an object when it is viewed from different lines of sight, in this case, both your eyes. Nearby objects have a larger parallax than further ones,

hence parallax can also be used to determine distance i.e. convey depth. Here the parallax barrier itself that consists of a thin layer of material with a series of slits cut into it. These slits are cut with great precision allowing each eye to see a different set of pixels, thereby creating a sense of depth. Once again, the problem of user positioning crops up. It is important to have a well-defined spot where the user will be else the 3D effect is lost.

Philips is one of the only companies that had an auto-stereoscopic display

range on the market, called WOWvx that allows 2D and 3D viewing. Although this series was mainly for commercial purposes and intended for advertising and marketing purposes, Philips themselves claim to have withdrawn it from the market sometime last year, in order to prevent another round of format wars. Their offering supported a resolution of 2160p – double of 1080p. 3D Auto-stereoscopic displays are still largely a work-in-progress, but there have been several prototypes, although there is nothing available in retail as the technology still needs a lot of work before it becomes market ready.

Holographic displays: the future

The concept has been around for a while. Anyone who has associated with the Star Wars series of movies and games will be familiar with the concept. Holography, in the realm of 3D displays, is based on the concept of “motion parallax” where the

perspective of the viewer can be variable and the image changes and moves continuously in keeping with the viewers' perspective. It is possible to view the object on screen from different directions, and each time new facets are visible, while some disappear – similarly to what happens in real life, when you walk around a solid body. The picture is not restricted to the screen itself – objects can appear to be both behind and in front of the screen. This is completely natural, and there is no motion tracking technology applied, like in lenticular lenses.

Such displays are made up of pixels, but these are referred to as voxels – or simply volumetric pixels, since instead of being represented by two coordinates on-screen, they are referred to by three coordinates i.e. a grid in 3 dimensional space. The concept of a voxel originated with holographic 3D displays. A voxel is, however, a lot more relative than a pixel, because its location is relative to the other voxels in a scene, since we're talking about a complex volumetric image that differs in composition as viewing orientation changes.

Holographic displays exist, but they are very application specific. The most obvious uses are medical and engineering applications, certain research also benefits from being able to view objects in 3D space. When it comes to home entertainment, holography is still a reasonable distance away – after all movies would need to be shot so as to have multiple orientation points – bandwidth requirements would be astronomical, as would the cost. Holographic images also persist for a while, making them unsuitable for any sort of rapid moving scenes – definitely not what you're looking for, for any sort of video watching. The technology ecosystem for such displays is also not proven, at least outside of a research lab – multimedia is a living, breathing medium and holographic displays are still too raw a technology to seem feasible. There have been breakthroughs in recent years – enough to lighten up prediction gurus' eyes, to the point where they believe we could see a 3D holographic display become feasible in the next 20 years. Wait and watch...**d**